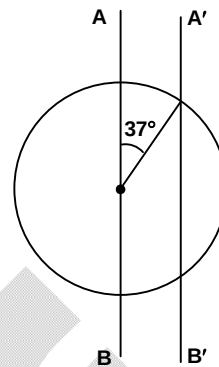


7. A ring of radius 2 m is kept on a smooth horizontal surface. A rod of sufficient length can slide over the ring as shown. Initially rod is kept along the diameter of the ring. Now, a velocity of magnitude 12 m/s is given to rod towards right and 8 m/s to ring towards left. After some time rod reaches at position A'B'. Choose the correct option(s) for position of A'B'.
- (A) Magnitude of velocity of point of intersection relative to ring is 25 m/s.
- (B) Magnitude of acceleration of point of intersection is $\frac{3125}{8} \text{ m/s}^2$.
- (C) Ratio of tangential acceleration to radial acceleration of point of intersection is 3/4.
- (D) Magnitude of tangential acceleration is $\frac{1875}{8} \text{ m/s}^2$.

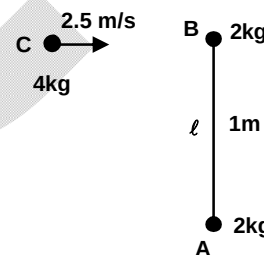


SECTION – B

(Numerical Answer Type)

This section contains **SIX (06)** Numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.

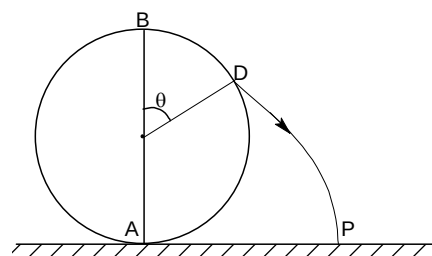
8. Two particles A and B each of mass 2 kg are connected by a massless rod of length $\ell = 1\text{m}$. Another particle C of mass 4 kg moving with a velocity 2.5 m/s (perpendicular to the rod AB) collides with the particle B as shown in the figure. Coefficient of restitution, $e = \frac{1}{2}$ and the system is placed in a gravity free space. The angular velocity (in rad/s) of the rod just after collision is k rad/s. Find the value of 10k.



9. A conducting square loop of side 2.0 cm is rotated through 180° about one of its diagonals in 0.20 s. A uniform magnetic field exists in the region which is perpendicular to the plane of the loop initially. If average induced emf is 20 mV, find the magnitude of the magnetic field (in Tesla).

10. A particle slides down the surface of a smooth fixed sphere of radius R starting from rest at the highest point B. Particle leaves the sphere at some point and then strikes the horizontal plane passing through the lowest point A of the sphere, at point P. The distance AP is given by

$$AP = \frac{n}{27} [\sqrt{n} + 4\sqrt{2}] R \text{ where 'n' is an integer, find n. (Take } g = 10 \text{ m/s}^2\text{).}$$



11. In the AC circuit shown in the figure, the rms current through the source is 2A. Find the value of capacitance C (in μF).

